In [2]:
```
import pandas as pd
import matplotlib.pyplot as plt
import numpy as np
import seaborn as sns
dataset=sns.load_dataset('titanic')
```

In [3]:
```
dataset.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 15 columns):
 #   Column       Non-Null Count  Dtype
---  ------       --------------  -----
 0   survived     891 non-null    int64
 1   pclass       891 non-null    int64
 2   sex          891 non-null    object
 3   age          714 non-null    float64
 4   sibsp        891 non-null    int64
 5   parch        891 non-null    int64
 6   fare         891 non-null    float64
 7   embarked     889 non-null    object
 8   class        891 non-null    category
 9   who          891 non-null    object
 10  adult_male   891 non-null    bool
 11  deck         203 non-null    category
 12  embark_town  889 non-null    object
 13  alive        891 non-null    object
 14  alone        891 non-null    bool
dtypes: bool(2), category(2), float64(2), int64(4), object(5)
memory usage: 80.7+ KB
```

In [4]:
```
dataset.head()
```

Out[4]:

	survived	pclass	sex	age	sibsp	parch	fare	embarked	class	who	adult_m
0	0	3	male	22.0	1	0	7.2500	S	Third	man	T
1	1	1	female	38.0	1	0	71.2833	C	First	woman	Fa
2	1	3	female	26.0	0	0	7.9250	S	Third	woman	Fa
3	1	1	female	35.0	1	0	53.1000	S	First	woman	Fa
4	0	3	male	35.0	0	0	8.0500	S	Third	man	T

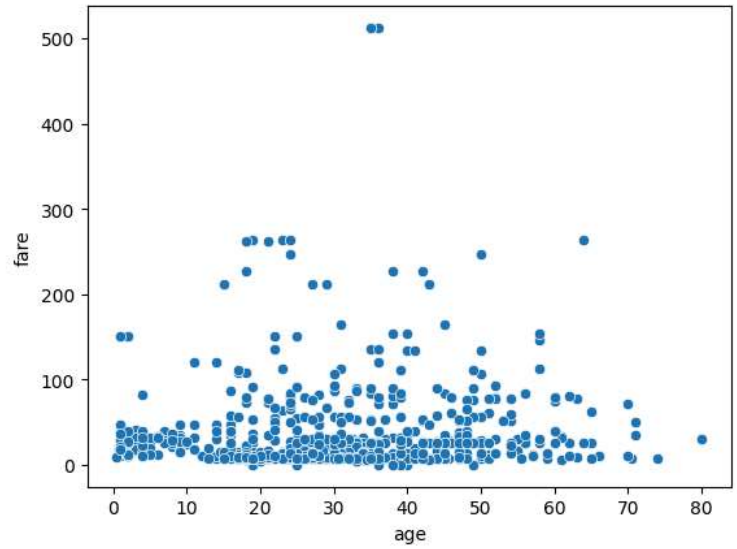
In [5]:
```
dataset.tail()
```

Out[5]:

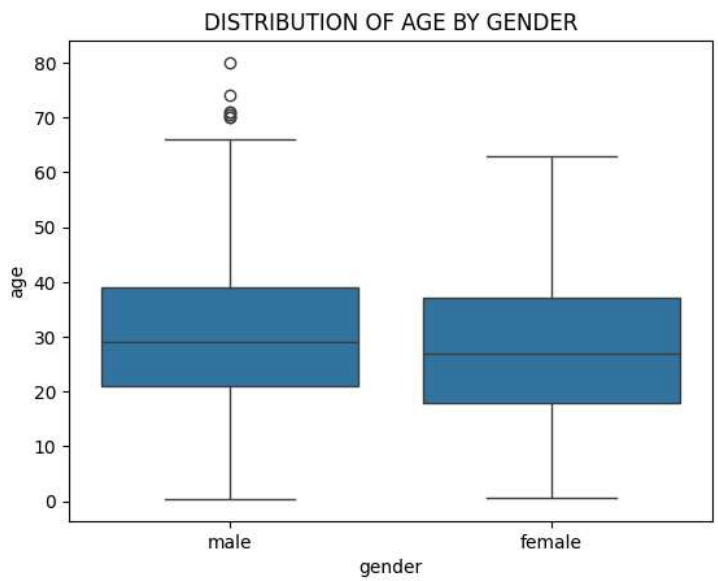
	survived	pclass	sex	age	sibsp	parch	fare	embarked	class	who	adult
886	0	2	male	27.0	0	0	13.00	S	Second	man	
887	1	1	female	19.0	0	0	30.00	S	First	woman	
888	0	3	female	NaN	1	2	23.45	S	Third	woman	
889	1	1	male	26.0	0	0	30.00	C	First	man	
890	0	3	male	32.0	0	0	7.75	Q	Third	man	

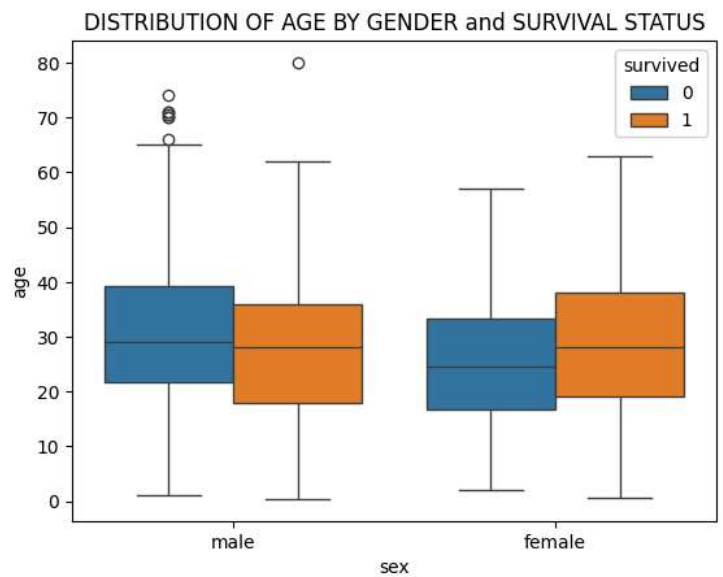
In [6]:
```
sns.scatterplot(data=dataset,x='age',y='fare')
```

Out[6]:
```
<Axes: xlabel='age', ylabel='fare'>
```



In [7]:
```
sns.boxplot(x='sex',y='age',data=dataset)  #median of male is 28 snd female is 29
plt.title('DISTRIBUTION OF AGE BY GENDER')
plt.xlabel('gender')
plt.ylabel('age')
plt.show()
```



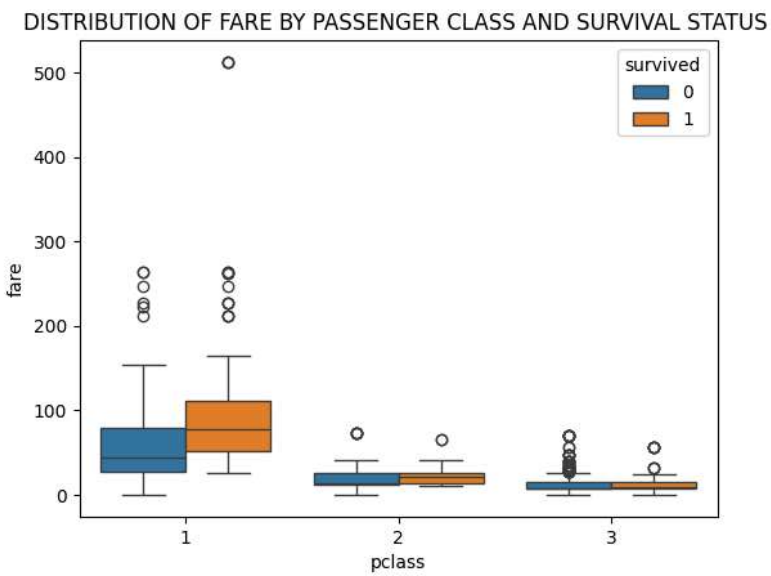


In [8]:
```
#boxplot of survived and not survived
sns.boxplot(x='sex',y='age',data=dataset,hue='survived')  #median of male is 28 snd
plt.title('DISTRIBUTION OF AGE BY GENDER and SURVIVAL STATUS')
```

Out[8]:
```
Text(0.5, 1.0, 'DISTRIBUTION OF AGE BY GENDER and SURVIVAL STATUS')
```

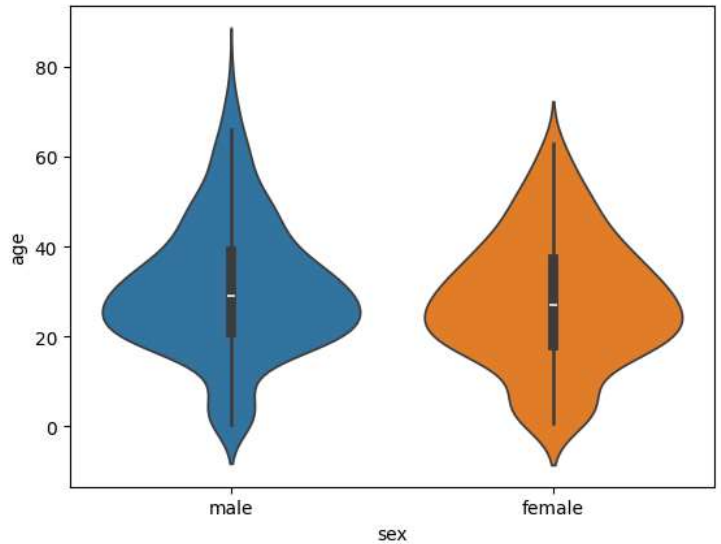
In [9]:
```
plt.title("DISTRIBUTION OF FARE BY PASSENGER CLASS AND SURVIVAL STATUS ")
plt.xlabel("pclass")
plt.ylabel("fare")
sns.boxplot(x='pclass',y='fare',data=dataset,hue='survived')
#those who have survived have paid more fare
```

Out[9]:
```
<Axes: title={'center': 'DISTRIBUTION OF FARE BY PASSENGER CLASS AND SURVIVAL STAT
US '}, xlabel='pclass', ylabel='fare'>
```



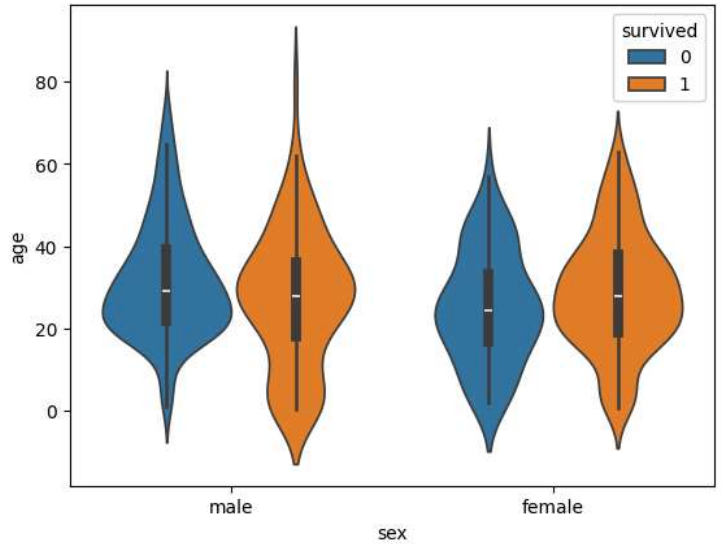
In [11]:
```
#violin plot
sns.violinplot(x='sex',y='age',data=dataset,hue='sex')
```

Out[11]: <Axes: xlabel='sex', ylabel='age'>



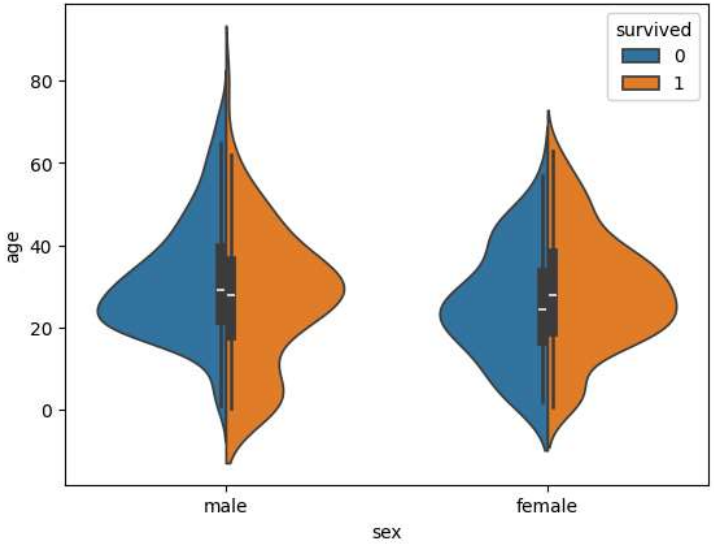
In [13]:
```
sns.violinplot(x='sex',y='age',data=dataset,hue='survived')
```

Out[13]: <Axes: xlabel='sex', ylabel='age'>



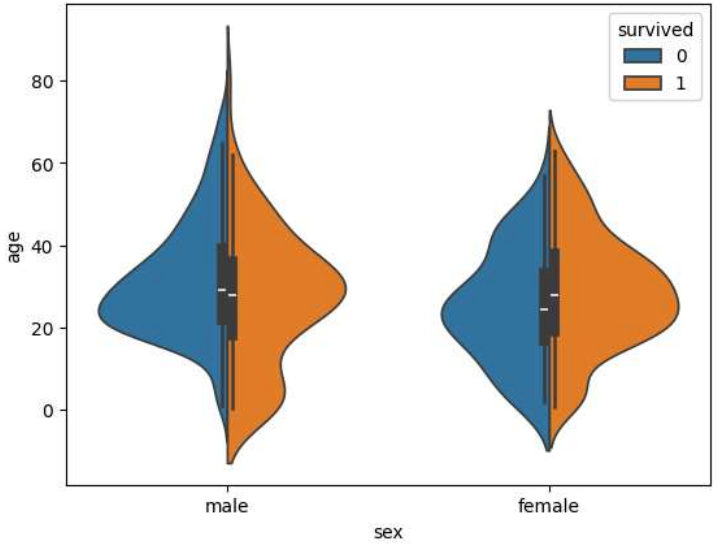
In [15]:
```
sns.violinplot(x='sex',y='age',data=dataset,hue='survived',split=True)
```

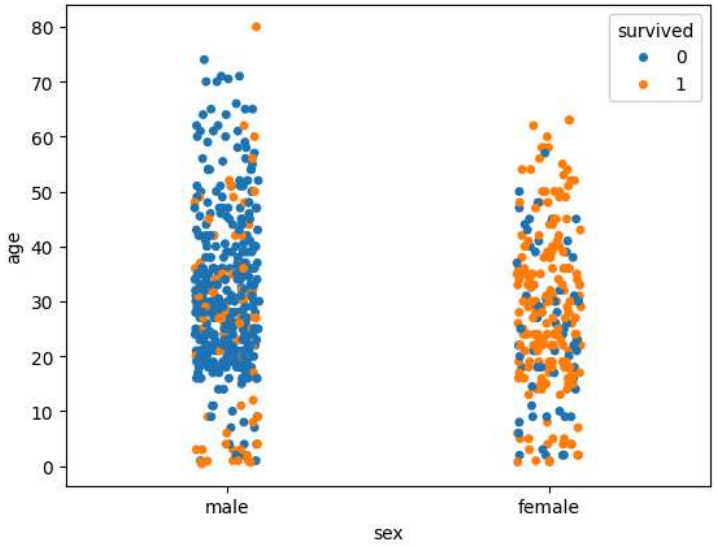
Out[15]: <Axes: xlabel='sex', ylabel='age'>



In [16]:
```
sns.violinplot(x='sex',y='age',data=dataset,hue='survived',split=True)
```

Out[16]: <Axes: xlabel='sex', ylabel='age'>



In [20]:
```
sns.stripplot(x='sex',y='age',data=dataset,hue='survived')
```

Out[20]: <Axes: xlabel='sex', ylabel='age'>